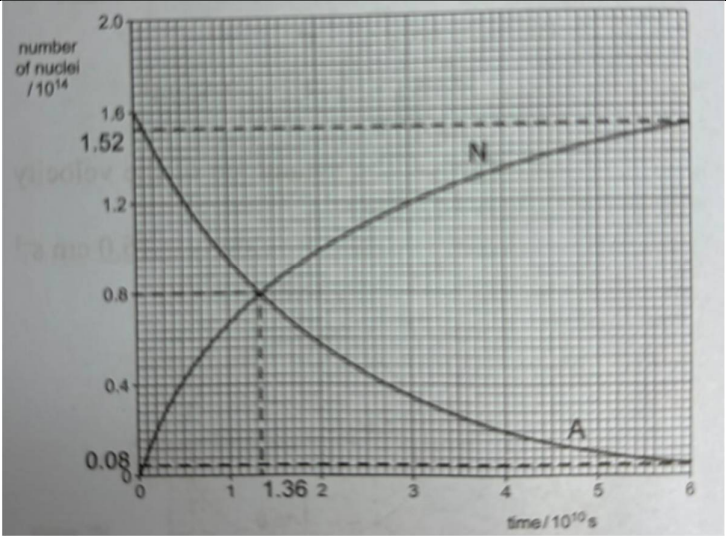


	For k that is halved, the period of oscillation would increase by a factor of $\sqrt{2}$.
9(a)(i)	$\Delta m = m_{\text{americium}} - m_{\text{neptunium}} - m_{\alpha}$ $= (400.19774 - 393.54304 - 6.64466) \times 10^{-27}$ $= 1.004 \times 10^{-27} \text{ kg}$ Energy released $= \Delta mc^2$ $= 1.004 \times 10^{-27} \times (3.0 \times 10^8)^2$ $= 9.036 \times 10^{-13}$ $= 9.0 \times 10^{-13} \text{ J (shown)}$
9(a)(ii)	The total momentum of a system is constant, provided no external resultant force acts on it.
9(a)(iii)	$E_k = \frac{1}{2}mv^2$ $E_k = \frac{1}{2} \frac{m^2 v^2}{m} = \frac{p^2}{2m}$ $p = \sqrt{2mE_k}$ $= \sqrt{2(6.64466 \times 10^{-27})(8.784 \times 10^{-13})}$ $= 1.080432 \times 10^{-19} \text{ kg m s}^{-1}$ Since the americium-241 nucleus was initially at rest, by principle of conservation of momentum, the momentum of the α-particle must be equal and opposite to the momentum of the neptunium-237 nucleus. Thus, $p_{\text{neptunium}} = 1.080432 \times 10^{-19} \text{ kg m s}^{-1}$ $p = \sqrt{2mE_k}$ $E_k = \frac{p^2}{2m}$ $= \frac{(1.080432 \times 10^{-19})^2}{2(393.54304 \times 10^{-27})}$ $= 1.48311 \times 10^{-14}$ $= 1.483 \times 10^{-14} \text{ J (shown)}$

Qn	Solution
9(a)(iv)	Most of the energy released in the decay in (a)(i) is converted to kinetic energy of the α -particle and neptunium-237 nucleus. However, a very small amount of the energy released is converted to the energy of a gamma ray photon that is also produced in the decay.
9(b)(i)	$A = \lambda N$ $N = \frac{A}{\lambda} = \frac{8.13 \times 10^3}{5.08 \times 10^{-11}} = 1.60 \times 10^{14}$
9(b)(ii)	$\lambda = \frac{\ln 2}{T_{\frac{1}{2}}} = \frac{\ln 2}{5.08 \times 10^{-11}} = 1.36 \times 10^{10} \text{ s}^{-1}$
9(b)(iii), (iv)	
9(c)(i)	The decay constant is defined as the probability of a radioactive nucleus decaying per unit time, so the decay constant of the beta emitter is much greater than the decay constant of americium-241. Since $A = \lambda N$ and both samples have the same initial number of nuclei, the initial activity of the sample of beta emitter is much higher.
9(c)(ii)	The time taken will be shorter for the beta emitter. The decay constant of the beta emitter is much greater than the decay constant of americium-241. Since $\lambda = \frac{\ln 2}{T_{\frac{1}{2}}}$, $T_{\frac{1}{2}}$ will be inversely proportional to decay constant, thus half life of beta emitter will be much shorter than half life of americium-241. Half life is defined as the expected time for activity of sample to halve, thus activity of beta emitter will take a shorter time to fall to 1 Bq.